

Identification Conditions for the CET in an ADL with Unit and Time Fixed Effects

Technical derivation for the IVB paper

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1 Purpose and target

This note gives conditions under which the population coefficient on current treatment in an autoregressive distributed-lag model with unit and time fixed effects equals the contemporaneous treatment effect (CET). It then separates that population identification result from finite- T bias of the within estimator and from inference under dependent errors. The target follows the within-period clock in the paper: the observed history is fixed, current treatment begins, within-period responses may occur, and the outcome is measured after the relevant exposure window.

For unit i in period t , let H_{it}^- denote the observed information and state variables fixed immediately before the first exposure represented by D_{it} . A common calendar-lag vector is

$$H_{i,t-1} = (Y_{i,t-1}, \dots, Y_{i,t-p}, D_{i,t-1}, \dots, D_{i,t-q}, X_{i,t-1}, \dots, X_{i,t-r})$$

where the number of lags is not restricted to one. Calendar-lag notation is a valid implementation of the causal history only when the timestamp checklist establishes both $H_{i,t-1} \subseteq H_{it}^-$ and that the recorded lag vector is sufficient for current assignment and the conditional outcome mean. A lag label alone establishes neither condition. Let $R_{it} = r(H_{it}^-)$ collect the transformations of the pre-exposure history included in the regression; under a valid calendar-lag implementation, these transformations can be written as $r(H_{i,t-1})$. For treatment levels d and d' in the relevant conditional support, define

$$\tau_{\text{CET}}(d, d'; h, a, t) = \mathbb{E}[Y_{it}(d) - Y_{it}(d') \mid H_{it}^- = h, \alpha_i = a, t].$$

Here $Y_{it}(d)$ sets current treatment to d while holding the realized pretreatment history fixed. Unless a controlled direct effect is explicitly targeted, $Y_{it}(d)$ includes causal paths through variables that respond after D_{it} within period t .

2 Assumptions

The following conditions are sufficient for the constant-coefficient result below. They are scope conditions, not consequences of estimating an ADL.

1. **Consistency and no interference.** If $D_{it} = d$, then $Y_{it} = Y_{it}(d)$. Unit i 's contemporaneous potential outcome does not depend on other units' treatments, and treatment versions are well defined.
2. **Observed pre-exposure history and no anticipation.** H_{it}^- is fixed before the first exposure relevant to D_{it} and contains the observed state needed to render current treatment conditionally as-if assigned. Anticipatory responses to D_{it} are absent. A calendar-lag vector $H_{i,t-1}$ can implement this condition only when its components are contained in H_{it}^- and are sufficient; coarse calendar labels do not establish either requirement.
3. **Sequential mean exchangeability.** For every supported d ,

$$\mathbb{E}[Y_{it}(d) \mid D_{it}, H_{it}^-, \alpha_i, \tau_t] = \mathbb{E}[Y_{it}(d) \mid H_{it}^-, \alpha_i, \tau_t].$$

Together with consistency and the linear conditional-mean restriction in Condition 5, sequential exchangeability implies $\mathbb{E}[u_{it} \mid D_{it}, H_{it}^-, \alpha_i, \tau_t] = 0$. The residual condition is therefore a consequence of the causal and functional-form restrictions, not an equivalent restatement of sequential exchangeability. Thus there is no unobserved contemporaneous cause of both D_{it} and Y_{it} after conditioning on the pre-exposure history, unit effect, and common time shock.

4. **Recursive within-period timing.** D_{it} is determined before the contemporaneous outcome innovation. There is no simultaneous feedback $Y_{it} \rightarrow D_{it}$ within the period represented by the model. Feedback $Y_{i,t-1} \rightarrow D_{it}$ is compatible with identification when $Y_{i,t-1} \in H_{it}^-$ and the conditioned history is sufficient; it is not a violation of sequential exchangeability. The problems are contemporaneous feedback or omission of the relevant pre-exposure state.
5. **Correct dynamic conditional mean.** The included lag structure and transformations span the relevant conditional mean, and the response is additive and linear:

$$\mathbb{E}[Y_{it}(d) \mid H_{it}^-, \alpha_i, \tau_t] = \alpha_i + \tau_t + R'_{it}\gamma + \beta_{\text{CET}}d. \quad (1)$$

In particular, omitted lags, nonlinear history terms, unit-specific trends, or latent time-varying states cannot be absorbed merely by labeling the model "ADL + FE."

6. **Homogeneity for a single coefficient.** The slope β_{CET} in (1) is constant over units, periods, histories, and the treatment contrasts being summarized. If effects or dynamics are heterogeneous, the regression coefficient generally becomes a projection-weighted summary rather than a common CET.
7. **Conditional positivity and overlap.** For every (h, a, t) where the CET is interpreted, both d and d' belong to the conditional support of D_{it} given $H_{it}^- = h$, $\alpha_i = a$, and t . For a binary or other discrete treatment, the relevant levels have positive conditional probability. For a continuous treatment, the target doses lie in the conditional common support and have local support, so the comparison does not extrapolate across unsupported dose regions. This causal support condition is not a rank condition for a linear projection.
8. **Residual treatment variance and projection rank.** If C_{it} collects R_{it} and the unit and time indicators and $\tilde{D}_{it} = D_{it} - P_C D_{it}$, then

$$0 < \mathbb{E}[\tilde{D}_{it}^2] < \infty. \quad (2)$$

Hence the population FWL coefficient is defined and treatment varies after the included history and fixed effects have been partialled out. This condition does not by itself establish conditional positivity for a causal contrast.

9. **Role of fixed effects and time shocks.** α_i absorbs additive time-invariant unit causes, and τ_t absorbs additive shocks common to all units in period t . These terms do not absorb unobserved time-varying unit-level confounding, unit-specific responses to common shocks, differential trends, or interference.

3 Population identification

Proposition 1 (Population CET and finite- T within estimation). *Suppose Conditions 1–9 hold and all variables have finite second moments. Define the population ADL + FE coefficient on D_{it} as the coefficient β_D^P in the linear projection of Y_{it} on D_{it} , R_{it} , unit indicators, and time indicators. Then*

$$\beta_D^P = \beta_{\text{CET}},$$

and, for every supported d, d' , $\tau_{\text{CET}}(d, d'; h, a, t) = \beta_{\text{CET}}(d - d')$.

This population equality does not imply that the conventional within estimator is unbiased at finite T . When R_{it} contains lagged outcomes, or when treatment is predetermined through observed past outcomes, the within transformation generally correlates transformed regressors with transformed innovations. The resulting Nickell-type estimation bias is a finite- T property of the estimator, not a failure of the population identification result. Serial or cross-sectional dependence in mean-zero innovations is a third issue: it changes valid uncertainty calculations but, by itself, does not redefine β_D^P .

3.1 Proof

Step 1: From the causal assumptions to an observed conditional mean. By consistency, observed outcomes equal the potential outcome under the realized current treatment. By sequential mean exchangeability,

$$\begin{aligned} \mathbb{E}[Y_{it} \mid D_{it} = d, H_{it}^-, \alpha_i, \tau_t] &= \mathbb{E}[Y_{it}(d) \mid D_{it} = d, H_{it}^-, \alpha_i, \tau_t] \\ &= \mathbb{E}[Y_{it}(d) \mid H_{it}^-, \alpha_i, \tau_t]. \end{aligned} \tag{3}$$

No anticipation makes the conditioning history a pretreatment state for the current intervention. Recursive timing ensures that the conditional comparison is not generated by a simultaneous outcome-to-treatment system. Substituting (1) into (3) gives

$$Y_{it} = \alpha_i + \tau_t + R'_{it}\gamma + \beta_{\text{CET}}D_{it} + u_{it}, \quad \mathbb{E}[u_{it} \mid D_{it}, H_{it}^-, \alpha_i, \tau_t] = 0. \tag{4}$$

Taking the difference between (1) evaluated at d and d' proves

$$\tau_{\text{CET}}(d, d'; h, a, t) = \beta_{\text{CET}}(d - d'). \tag{5}$$

Step 2: The population projection coefficient. Let C_{it} contain R_{it} and the unit and time indicators, and let $\tilde{D}_{it} = D_{it} - P_C D_{it}$ be the population residual from projecting treatment on C_{it} . The population Frisch–Waugh–Lovell coefficient is

$$\beta_D^P = \frac{\mathbb{E}[\tilde{D}_{it}Y_{it}]}{\mathbb{E}[\tilde{D}_{it}^2]}. \quad (6)$$

Substitution of (4), orthogonality of $\tilde{D}_{it}$ to C_{it} , and conditional mean zero yield

$$\mathbb{E}[\tilde{D}_{it}Y_{it}] = \beta_{\text{CET}}\mathbb{E}[\tilde{D}_{it}^2] + \mathbb{E}[\tilde{D}_{it}u_{it}] = \beta_{\text{CET}}\mathbb{E}[\tilde{D}_{it}^2]. \quad (7)$$

The residual-variance condition in (2) therefore implies $\beta_D^P = \beta_{\text{CET}}$. Conditional positivity in Condition 7 identifies the causal contrasts being summarized, while the distinct rank condition in (2) makes the population projection coefficient well defined. Unit and time fixed effects contribute to this result only by spanning the additive unit factors and common time shocks specified in (1); they do not make (7) true when the conditional mean omits relevant time-varying causes.

Step 3: What the coefficient means without linearity or homogeneity. If the conditional response is instead $\beta_{it}(H_{it}^-)D_{it}$, while the remaining conditional mean is correctly spanned, then

$$\beta_D^P = \frac{\mathbb{E}[\tilde{D}_{it}\beta_{it}(H_{it}^-)D_{it}]}{\mathbb{E}[\tilde{D}_{it}^2]}. \quad (8)$$

This expression need not equal an unweighted average CET and need not have convex weights. Nonlinear dose responses likewise make (6) a best linear approximation over the observed support. Heterogeneous treatment effects, heterogeneous lag coefficients, and misspecified lag order are therefore scope limitations, not nuisances removed automatically by fixed effects.

Step 4: Why the within estimator can be biased at finite T . Stack the regressors whose slopes are to be estimated in $X = [D, R]$, let F contain unit and time indicators, and let M_F be the residual-maker for those indicators. If $\vartheta = (\beta_{\text{CET}}, \gamma')'$, the conventional within estimator obeys the exact sample identity

$$\hat{\vartheta}_W - \vartheta = (X'M_F X)^{-1}X'M_F u. \quad (9)$$

The level restriction $\mathbb{E}[u_{it} \mid D_{it}, H_{it}^-, \alpha_i, \tau_t] = 0$ does not generally imply $\mathbb{E}[X'M_F u] = 0$ at fixed T . To see the source, suppress time effects for notation and write

$$\ddot{u}_{it} = u_{it} - \bar{u}_i, \quad \ddot{Y}_{i,t-1} = Y_{i,t-1} - \bar{Y}_{i,-1}. \quad (10)$$

The lagged outcome contains earlier innovations such as $u_{i,t-1}$, while $\bar{u}_i$ also contains those innovations. Consequently $\mathbb{E}[\ddot{Y}_{i,t-1}\ddot{u}_{it}] \neq 0$ in general, even when the level innovation is sequentially mean independent. If D_{it} responds to $Y_{i,t-1}$ or other past shocks, the transformed treatment can inherit the same finite- T correlation. Thus lagged feedback $Y_{i,t-1} \rightarrow D_{it}$ is compatible with the population argument in Steps 1–2 when $Y_{i,t-1} \in H_{it}^-$ and the conditioned history is sufficient, yet it can still matter for the finite- T behavior of the within estimator.

In the standard dynamic fixed-effects setting, this is the source of Nickell-type bias (Nickell, 1981). Its leading order is commonly $1/T$ under the usual stationary large- N , large- T regularity conditions, so increasing N with fixed T does not remove it. The exact magnitude and the induced bias in the

treatment coefficient depend on persistence, treatment assignment, the regressor covariance matrix, initial conditions, and the panel design. Equation (9) therefore separates the identified population coefficient from the estimator used to approximate it. This completes the proof. $\square$

4 Identification, estimation, and inference are different questions

Table 1: Identification, estimation, and inference for the ADL + FE benchmark

Layer	Question	Required conditions	What failure means
Population identification	Does the coefficient equal the CET?	Defended causal clock, consistency, no interference, sequential exchangeability given H_{it}^- and fixed effects, correct conditional mean, linearity, and homogeneity	The coefficient may reflect confounding, simultaneity, misspecified dynamics, or only a best linear approximation
Conditional support	Is the target causal contrast supported?	Positive conditional probability for discrete levels; local conditional common support for continuous doses	The contrast requires unsupported extrapolation even if a linear coefficient exists
Projection rank	Is the population FWL coefficient well defined?	$0 < \mathbb{E}[\tilde{D}_{it}^2] < \infty$ after projecting on the included history and fixed effects	The linear coefficient is not well defined even if some causal contrasts have overlap
Finite- T estimation	Does the within estimator recover the identified coefficient in the available panel?	Sufficient time dimension or an estimator/correction whose moments and asymptotics fit the design	Lagged outcomes and predetermined regressors can produce Nickell-type bias even when the population CET is identified
Inference	Are standard errors and confidence intervals valid?	An uncertainty procedure appropriate for serial dependence, cross-sectional dependence, clustering structure, and the relevant asymptotic sequence	Point identification need not change, but reported uncertainty can be wrong; a residual common factor correlated with treatment is instead an identification failure

Serial correlation in u_{it} and residual dependence across units affect the covariance of the estimator and the validity of standard errors. If dependence is mean zero conditional on the regressors, it is an inference problem rather than a change in the target defined by (6). Appropriate procedures depend on the sampling and dependence structure; unit clustering, two-way clustering, spatial or network corrections, and common-correlated-effects approaches impose different conditions. If an omitted common or unit-specific shock is correlated with D_{it} after conditioning, however, the issue returns to identification rather than inference.

5 Short panels and the GMM boundary

When T is short enough that Nickell-type bias is substantively important, bias-corrected fixed-effects estimators, split-panel or half-panel jackknife methods, or moment-based estimators may be considered. Arellano–Bond difference GMM removes the unit effect by differencing and uses lagged levels as instruments under additional moment restrictions (Arellano and Bond, 1991). It is an alternative estimator for a particular short- T , large- N design, not a universally superior version of FE and not evidence that the population assumptions above hold. Instrument validity, serial-correlation restrictions, persistence and weak instruments, instrument proliferation, finite-sample behavior, and the treatment’s own predetermined or endogenous status must all be defended.

The distinction mirrors the broader dynamic-panel literature: causal identification depends on the treatment history, timing, and counterfactual assumptions (Blackwell and Glynn, 2018; Imai and Kim, 2021); finite-panel corrections and GMM address properties of estimators under their own additional assumptions. Neither solves unobserved contemporaneous confounding, simultaneous treatment–outcome determination, interference, or misspecified causal timing.

6 Empirical scope conditions left to defend

The proposition is conditional. An application must still establish that the recorded treatment precedes the outcome innovation; that any calendar-lag vector lies within H_{it}^- and is a sufficient observed history; that anticipation and interference are absent; that remaining unit and time heterogeneity is additive; that the target contrast has conditional overlap; that treatment retains nonzero residual variance for the population projection; and that a constant linear CET is substantively meaningful. It must separately assess whether T is long enough for within-estimator bias to be negligible or corrected and whether the uncertainty procedure matches the serial and cross-sectional dependence in the data.

References

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